

Josh Lazaruk

Engineering Physics, BSc Co-op | Class of 2022 | University of Alberta, Canada

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Skills

Design Cycle: prototyping, agile scrum, system modeling, iterative testing, cross-functional leadership

Programming: C, C++, Python, Robot Framework, MATLAB, VHDL, Git, Linux, FreeRTOS

Software: Altium Designer, Simulink, STK, SPICE, PTC Windchill Modeler (SysML)

Hardware/Equipment: RF test equipment, oscilloscope, logic analyzer, multimeter, soldering, CPUs

Project Experience

Systems Team Lead - AlbertaSat

September 2019 - November 2022

- Oversaw system requirements and configuration control, applying NASA's systems engineering principles to ensure critical reliability
- Established documentation standards to ensure clear communication across engineering, research, and project teams
- Managed system-wide testing and integration of radio, power, attitude control, imaging, and computing subsystems of 3 CubeSats
- Made strategic decisions and provided governance for a 3-year project with an ~\$800 000 budget as one of six members of the AlbertaSat Steering Committee

Software Team Lead - AlbertaSat

May 2019 - August 2019

- Managed a team of 13 software developer volunteers, establishing onboarding materials
- Created work breakdown structure and Gantt chart for software subsystems of 3 CubeSats

Professional Experience

Test Specialist - Xiphos Systems Corporation

December 2022 - January 2025

- Designed and executed all qualification testing and requirements verification of two custom high-performance spacecraft computer system designs
- Improved production test coverage and reliability for 100s of spacecraft computers/yr
- Designed PCB and harnesses for high-speed production testing, optimizing 500+ Mbps data interfaces for scalability and reliability

Satellite Systems Integrator & Tester - AlbertaSat

May 2022 - November 2022

- Led assembly, integration and testing of three cube satellites and two ground stations
- Managed 12-person interdisciplinary team using agile scrum framework
- Developed two GNURadio software-defined radio systems, capable of 10 Mbps data rate
- Ensured adherence to external test and documentation standards, maintaining compliance and quality throughout satellite development

Systems Engineering Intern - Canadian Space Agency

May 2021 - August 2021

- Facilitated Canadarm3 requirements and system model development
- Developed program-wide Systems Engineering Management Plan for Canadarm3
- Improved procedures used for interchanging data with contractor
- Implemented hundreds of requirements modifications in DOORS database
- Learned processes and software tools used in a \$50M+ international project

Embedded Systems Designer - AlbertaSat

January 2020 - December 2020

- Developed the open-source Athena II onboard computer, successfully deployed on 3 CubeSats in 2022, enhancing mission data handling and reliability
- Performed full component selection, schematics design, and layout for computer PCB
- Assembled, programmed, and functionally tested PCBs with >250 components
- Programmed and tested I2C, SPI, CAN, and UART peripheral drivers

Presentations and Conference Papers

J. Cai et al. 2023. "Applications of Systems Engineering Methodologies to an Undergraduate Student Volunteer-Led CubeSat Constellation," Poster accepted for SmallSat Poster Session 1 at the 39th Annual Small Satellite Conference, Logan, UT.

J. Lazaruk et al. 2022. "Open-Source CubeSat Optical Communication Receiver," Poster accepted for SmallSat Poster Session 4 at the 36th Annual Small Satellite Conference, Logan, UT.

A. Hoover et al. 2021. "Northern SPIRIT Consortium - Canadian Collaboration through Student-led CubeSat Constellation," *Proceedings of the Small Satellite Conference*, Swifty Session 10: The Case for Space, SSC21-S1-43. <https://digitalcommons.usu.edu/smallsat/2021/all2021/119/>

J. Lazaruk, B. Dobberthien, A. Tessier, A. Yahya, Long echo time magnetic resonance spectroscopy for aspartate measurement at 9.4 T. COMP annual scientific meeting (virtual), Medical Physics V. 48, p. 4699:2021 (poster presentation). Erratum in: Medical Physics V. 48, p. 7564:2021.

Awards & Scholarships

NSERC Undergraduate Student Research Award, 2020
Jason Lang Scholarship for Academic Excellence, 2019
NSERC Undergraduate Student Research Award, 2019
Jason Lang Scholarship for Academic Excellence, 2018

Certifications & Training

Project Management Principles & Practices Level 1, UofA Technology Training Center, 2020
Amateur Radio Operator - Advanced Qualification, 2020
STK (orbital mechanics modeling and mission analysis software) - Level 1, 2019

Interests

Robotics, AI, space, virtual reality, music, cooking, disc golf, movies, and reading